

Project 01: Real-Time Text-to-SQL AI Studio

Natural Language Database Analytics Powered by Google Gemini 2.5 Flash

1. Executive Overview

The Text-to-SQL AI Studio translates natural language questions into safe, executable SQLite queries. It features a continuous real-time e-commerce transaction stream, interactive Schema Explorer, read-only SQL safety validator, CSV data exporter, and SSE streaming AI query explanations.

2. System Architecture & Workflow Pipeline

- User Question Input -> SchemaRetriever & Few-Shot Examples
- Google Gemini 2.5 Flash Generator (Fallback: Schema Rule Engine)
- SQLValidator (Regex & Read-Only Safety Verification)
- SQLite Execution Engine (timeout=30.0s) -> Tabular Results & Chart UI

3. API Endpoints Specification

Method	Endpoint	Description
GET	/	Serves single-page glassmorphism web application
GET	/api/schema	Returns database schema summary
GET	/api/realtime-stats	Returns live revenue, total order count, and latest transaction
POST	/api/query	Translates natural language to SQL and executes query
POST	/api/simulate-live-order	Triggers a real-time order transaction
GET	/api/stream-explain	Streams token-by-token AI query explanations via SSE

4. Deployment & Automated Verification

Standalone Uvicorn Launch:

```
cd 01_text_to_sql/src && python -m uvicorn server:app --host 127.0.0.1 --port 8001
```

Automated Test Suite Execution:

```
python -m pytest 01_text_to_sql/tests/
```